

Saunak Kumar Panda

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Research Interests

- Reinforcement learning and sequential decision-making under uncertainty
- Applied optimization and data analytics for manufacturing, healthcare, and electrical systems
- LLM-augmented decision support for engineering, business and healthcare applications

Education

University of Houston – PhD in Industrial Engineering, GPA: 3.96	Aug 2025
• Specialization: Operations Research	
Texas Tech University (Transferred) – PhD in Industrial Engineering, GPA: 4.0	Aug 2022
University of Washington – MS in Mechanical Engineering, GPA: 3.82	June 2020
• Specialization: Robotics	
PES Institute of Technology – BE in Mechanical Engineering, GPA: 3.7	June 2018
• Minor: Electronics and Communication	

Publications

Peer-Reviewed Publications

- **Saunak Kumar Panda**, Yisha Xiang, and Ruiqi Liu, "Dynamic resource matching in manufacturing using deep reinforcement learning".
In: *European Journal of Operational Research (EJOR)*.
URL: <https://doi.org/10.1016/j.ejor.2024.05.027>
- **Saunak Kumar Panda**, Yisha Xiang, and Ruiqi Liu, "Online Statistical Inference of time-varying sample-averaged Q-learning".
In: *IEEE Transactions on Information Theory (IEEE TIT)*.
URL: <https://doi.org/10.1109/TIT.2025.3569421>
- **Saunak Kumar Panda**, Ruiqi Liu, and Yisha Xiang, "Online Statistical Inference of Sample-averaged Q-Learning".
In: *First Reinforcement Learning Safety Workshop, Reinforcement Learning Conference (RLC 2024)*.
URL: <https://openreview.net/forum?id=bKXRTxxmey>

Working Papers / Under Review

- **Saunak Kumar Panda**, Tong Li, Yisha Xiang, and Ruiqi Liu, "Empirical Variance-Penalized Risk-Sensitive Reinforcement Learning".
Under Review: *IEEE Transactions on Neural Networks and Learning Systems (IEEE TNNLS)*.
- Tong Li, **Saunak Kumar Panda**, and Yisha Xiang, "Certifying Lower Bounds for Risk-Sensitive Reinforcement Learning under Adversarial State Perturbations".
Under Review: *Machine Learning*.
- **Saunak Kumar Panda**, Ying Lin, and Meng Li, "A Two-Stage Ranking Framework for ABA Treatment Goal Recommendation".
Preparation for Submission: *Journal of Biomedical Informatics (JBI)*.

- **Saunak Kumar Panda**, Jingyi Tian, and Meng Li, "Deep RL in Procurement Auctions".
Preparation for Submission: *Manufacturing & Service Operations Management (M&SOM)*.
- **Saunak Kumar Panda**, Masoud Heidary, Biresh Kumar Joardar, and Yisha Xiang, "Deep reinforcement learning for optimization of machine learning on manycore circuit design".
Preparation for Submission: *International Symposium on Quality Electronic Design (ISQED)*.

Research Experience

Postdoctoral Scholar, HAI Institute, University of Houston – Houston TX Sep 2025 – Present

- Conduct research in reinforcement learning, machine learning, and human-centered AI, with applications in healthcare operations, recommendation systems, and algorithmic decision-making.
- Develop data-driven methods that combine predictive modeling, optimization, and domain expertise to support complex real-world decisions.

Graduate Research Assistant, University of Houston – Houston TX Aug 2020 – Aug 2025

- Working on multiple projects in large-scale optimization intended for publication in reputable OR and optimization journals
- Research Assistant for Dr. Yisha Xiang

BARC Graduate Researcher, Boeing Advanced Research Center – Seattle, WA Jul 2019 – Sep 2019

- Developed an automated inspection technique for tube bending using computer vision and linear regression, achieving a 75% accuracy improvement and reducing bend angle errors to ± 0.1 degrees
- **Concepts:** Computer vision, Machine Learning
- **Technical Skills:** OpenCV, PyTorch, MATLAB

Precision Controls Researcher, Microsoft Research Labs – Redmond, WA Jan 2019 – Dec 2019

- Designed PID control algorithms for robotic eye-tracking systems, improving laser dot detection and calibration accuracy, and enhancing system robustness
- **Concepts:** PID Control, Computer vision
- **Technical Skills:** OpenCV, Python

Research Intern, Indian Institute of Technology (IIT) Bombay – Mumbai, India Jan 2017 – Jun 2017

- Achieved more efficient shape optimization of plates under various loading conditions by developing a novel variant of Particle Swarm Optimization (PSO)
- Conducted comprehensive benchmark comparisons between PSO and other contemporary optimization algorithms to evaluate performance improvements

Grants and Proposals

National Science Foundation (NSF) – Smart and Connected Health (SCH) Proposal in Preparation, 2026

Collaborative Research: Smart, Secure and Scalable (S3) Health Monitoring of Large Populations via Online Collaborative Learning

- Contributing to the Human-Centered Recommendation thrust with Dr. Ying Lin (PI), developing methods for interpretable, clinically deployable recommendation systems in Applied Behavior Analysis (ABA), with emphasis on explainable AI, clinician-facing decision support, and real-world experimental evaluation.

Teaching Experience

Instructor, Data Analysis and Visualization, HAI Institute, University of Houston Fall 2025

- Designed and taught an 11-session course (~60 students) on business and data analytics using Excel and Power BI.

Curriculum Developer, Decision Modeling and Optimization, University of Houston Spring 2023

- Developed reinforcement learning modules for Dr. Yisha Xiang's graduate course.

Teaching Assistant, Deterministic OR, Texas Tech University Fall 2021

- Assisted Dr. David Huckleberry Gutman with course delivery, grading, and problem-solving sessions in linear programming and network optimization.

Industry Experience

Systems Modeling & Simulation Intern, MicroVision Inc. – Redmond, WA Jan 2020 – Mar 2020

- Developed system-level models and low-level controls for automotive LiDAR systems using LTSpice, enhancing accuracy and robustness through comprehensive unit and functional testing
- Reduced model error and identified additional faults, ensuring reliable system deployment in real-world conditions

Project Intern, Robert Bosch Engineering and Business Solutions – Bangalore, India Jan 2018 – Apr 2018

- Developed an Excel tool for material quantity and price calculations, optimizing cost estimation for additive manufacturing
- Collaborated with the marketing team to design a brochure for the Additive Manufacturing division, improving communication of technical capabilities

Recognition and Awards

Finalist - INFORMS Railway Applications Section (RAS) 2025 Data Challenge

- Recognized among the top six international teams for developing an end-to-end predictive maintenance pipeline for railway wheel failure prediction.

Distinguished Scholarship Award - University of Houston

- Awarded for exceptional research contributions and academic performance in operations research.

Skills

Programming Languages: Python, R, MATLAB, C, C++, Java, SQL, Linux

Libraries & Frameworks: PyTorch, TensorFlow, Scikit-learn, OpenAI Gym, OpenCV, ROS, Pandas, NumPy, JAX, Git

Solvers & Software: Gurobi, CPLEX, Simulink, LabVIEW, SolidWorks, Ansys, COMSOL, PLC, HMI

Invited & Contributed Talks

2025 INFORMS Annual Meeting <i>RAS Data Challenge - Predictive Maintenance in Rail Operations: Railway Wheel Failure Prediction</i> <i>Empirical Variance-Penalized Risk-Sensitive Reinforcement Learning</i>	Oct 2025
2024 INFORMS Annual Meeting Online Statistical Inference of time-varying sample-averaged Q-learning	Oct 2024
Reinforcement Learning Conference (RLC) 2024 Online Statistical Inference of sample-averaged Q-learning	Aug 2024
2023 INFORMS Annual Meeting Online Statistical Inference for dynamically-changing batch Q-learning	Oct 2023
University of Houston Industrial Engineering (UHIE) Friday Seminar Series Solving Dynamic Resource Matching in Manufacturing using Reinforcement Learning	Oct 2022
2022 INFORMS Annual Meeting Solving Dynamic Resource Matching in Manufacturing using Reinforcement Learning	Oct 2022
Institute of Industrial and Systems Engineers Annual Conference & Expo 2022 Dynamic matching of demand-supply types with manufacturing resources	May 2022

Services

President, INFORMS Student Chapter, University of Houston - Led a student chapter of over 20 members, promoting academic excellence and advancing operations research through coordinated efforts. - Organized seminars, facilitated weekly meetings, and developed materials for social and academic events to foster a cohesive and collaborative research community.	Aug 2022 – Aug 2025
Student Board Member, IISE QCRE Edited and published the quarterly newsletter, ensuring timely and accurate communication of chapter updates, events, and achievements.	Aug 2022 – May 2024
Conference Session Chair <i>2023 INFORMS Annual Meeting: Advances in Reinforcement Learning</i> <i>IISE Annual Conference & Expo 2022: Operations Research</i>	2023 2022